

National University of Computer and Emerging Sciences



Peshawar Campus

**Programming Fundamentals
Assignment-1**

Submitted By

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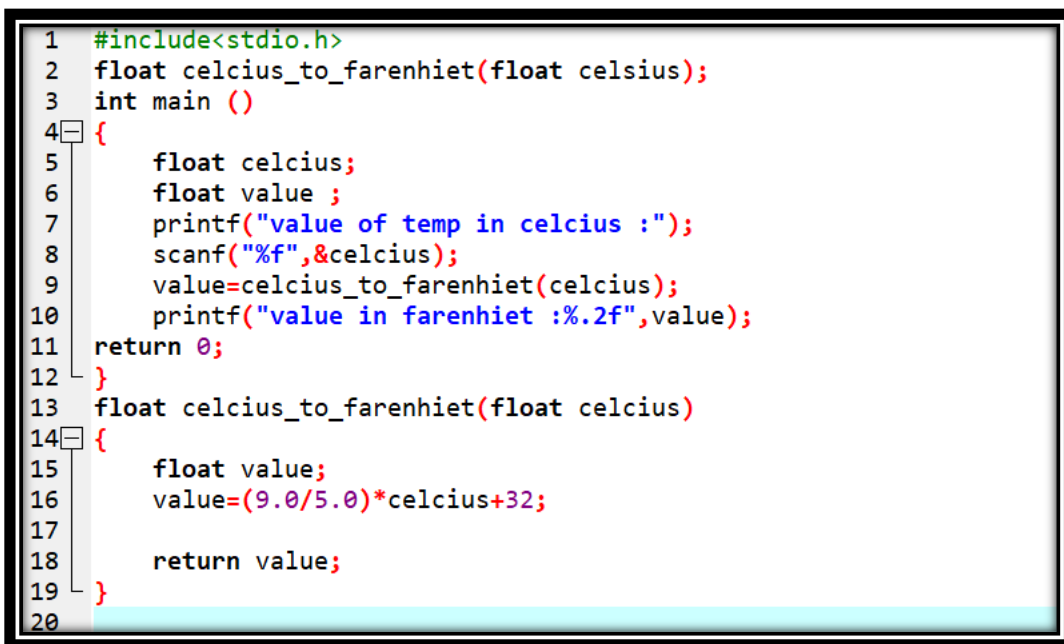
Submitted To

Sir Usman

Problem01

```
#include<stdio.h>
float celcius_to_farenhiet(float celsius);
int main ()
{
    float celcius;
    float value ;
    printf("value of temp in celcius :");
    scanf("%f",&celcius);
    value=celcius_to_farenhiet(celcius);
    printf("value in farenhiet :%.2f",value);
return 0;
}
float celcius_to_farenhiet(float celcius)
{
    float value;
    value=(9.0/5.0)*celcius+32;

    return value;
}
```

A screenshot of a code editor with a black border. The code is written in C and is color-coded: keywords like 'include', 'float', 'int', 'return', and 'scanf' are in red; string literals are in blue; and mathematical expressions and variables are in black. Line numbers 1 through 20 are visible on the left side of the editor. The code is identical to the one in the previous block.

```
1  #include<stdio.h>
2  float celcius_to_farenhiet(float celsius);
3  int main ()
4  {
5      float celcius;
6      float value ;
7      printf("value of temp in celcius :");
8      scanf("%f",&celcius);
9      value=celcius_to_farenhiet(celcius);
10     printf("value in farenhiet :%.2f",value);
11     return 0;
12 }
13 float celcius_to_farenhiet(float celcius)
14 {
15     float value;
16     value=(9.0/5.0)*celcius+32;
17
18     return value;
19 }
20
```

PROBLEM02

```
#include<stdio.h>
int year_is_leap(int);
int main()
{
    int year;
    printf("year :");
    scanf("%d",&year);
    year_is_leap(year);
}
int year_is_leap(int year)
{
    if(year%400==0 || year%4==0 && year%100!=0)
        printf("year is leap :");
    else
        printf(" year is not leap year :");
    return 0; }
```

```

1  #include<stdio.h>
2  int  year_is_leap(int);
3  int main()
4  {
5      int year;
6      printf("year :");
7      scanf("%d",&year);
8      year_is_leap(year);
9  }
10 int year_is_leap(int year)
11 {
12     if(year%400==0 || year%4==0 && year%100!=0)
13         printf("year is leap :");
14     else
15         printf(" year is not leap year :");
16     return 0; }
17

```

PROBLEM 03

```
#include<stdio.h>
```

```

    int BottleOfCookie(int x,int y ,int z){
//Number of Boxes, Number of Containers to Ship
        int perBoxCookies = x/y;
        int WastedBox = x%y;
printf("Boxex Ready to be Shiped : %d \n Discarded Box : %d \n
        ",perBoxCookies,WastedBox);

        int TotalContainer = perBoxCookies/z;
        int ContainerNotReady = perBoxCookies%z;
return printf("Container Ready to be Shiped : %d \n Discarded
        Container %d \n",TotalContainer,ContainerNotReady);

    }

```

```

int main(){
    int TotalCookie, BoxCookie, ContainerCookie;
    printf("Enter the Total Number of Cookies Available : ",TotalCookie);
        scanf("%d",&TotalCookie);
    printf("Enter the Number of Cookies in Box Available : ",BoxCookie);
        scanf("%d",&BoxCookie);
    printf("Enter the Number of Cookie Boxes in a Container Available : 
        ",ContainerCookie);
        scanf("%d",&ContainerCookie);
    printf("%d",BottleOfCookie(TotalCookie,BoxCookie,ContainerCookie));

    return 0;

}

```

```

#include<stdio.h>

int BottleOfCookie(int x,int y ,int z){
    //Number of Boxes, Number of Containers to Ship
    int perBoxCookies = x/y;
    int WastedBox = x%y;
    printf("Boxes Ready to be Shipped : %d \n Discarded Box : %d \n ",perBoxCookies,WastedBox);
    int TotalContainer = perBoxCookies/z;
    int ContainerNotReady = perBoxCookies%z;
    return printf("Container Ready to be Shipped : %d \n Discarded Container %d \n",TotalContainer,ContainerNotReady);
}

int main(){
    int TotalCookie, BoxCookie, ContainerCookie;
    printf("Enter the Total Number of Cookies Available : ",TotalCookie);
    scanf("%d",&TotalCookie);
    printf("Enter the Number of Cookies in Box Available : ",BoxCookie);
    scanf("%d",&BoxCookie);
    printf("Enter the Number of Cookie Boxes in a Container Available : ",ContainerCookie);
    scanf("%d",&ContainerCookie);
    printf("%d",BottleOfCookie(TotalCookie,BoxCookie,ContainerCookie));
    return 0;
}

```

```
C:\Users\Lenovo\Desktop\1.exe
Enter the Total Number of Cookies Available : 15
Enter the Number of Cookies in Box Available : 2
Enter the Number of Cookie Boxes in a Container Available : 6
Boxes Ready to be Shipped : 7
  Discarded Box : 1
  Container Ready to be Shipped : 1
  Discarded Container 1
58
-----
Process exited after 6.11 seconds with return value 0
Press any key to continue . . .
```

PROBLEM 04

```
#include<stdio.h>

int main()
{
    float x;
    printf("magnitude of earthquake");
    scanf("%f",&x);
    if (x<2.0)
        printf("micro");
    if(x>=2.0&&x<3.0)
        printf("x very minor");
    if(x>=3.0&&x<4.0)
        printf("minor");
    if(x>=4.0&&x<5.0)
```

```
printf("light");  
if(x>=5.0&& x<6.0)  
printf("moderate");  
if(x>=6.0&& x<7.0)  
printf("strong");  
if(x>=7.0&& x<8.0)  
printf("major");  
if(x>=8.0&& x<10.0)  
printf("great");  
if(x>=10.0)  
printf("meteoric");  
return 0;
```

```
#include<stdio.h>  
int main()  
{  
    float x;  
    printf("magnitude of earthquake");  
    scanf("%f",&x);  
    if (x<2.0)  
        printf("micro");  
    if(x>=2.0&&x<3.0)  
        printf("x very minor");  
    if(x>=3.0&&x<4.0)  
        printf("minor");  
    if(x>=4.0&&x<5.0)  
        printf("light");  
    if(x>=5.0&&x<6.0)  
        printf("moderate");  
    if(x>=6.0&&x<7.0)  
        printf("strong");  
    if(x>=7.0&&x<8.0)  
        printf("major");  
    if(x>=8.0&&x<10.0)  
        printf("great");  
    if(x>=10.0)  
        printf("meteoric");  
    return 0; |  
}
```

C:\Users\Lenovo\Desktop\1.exe

magnitude of earthquake 5
moderate

Process exited after 5.52 seconds with return value 0
Press any key to continue . . .

PROBLEM 05

```
#include<stdio.h>

int main ()
{
    int x, y, z;
    printf("first number");
    scanf("%d",&x);
    printf("second number");
    scanf("%d",&y);
    printf("third number");
    scanf("%d",&z);
```

```

        if(x>y && x>z){
printf("%d is the largest ODD value mong three digit \n",x);
        }
        if(y>x && y>z){
printf("%d is the largest ODD value mong three digit \n",y);
        }
        if(z>y && z>x){
printf("%d is the largest ODD value mong three digit \n",z);
        }
        return 0;
    }

```

```

#include<stdio.h>
int main ()
{
    int x, y, z;
    printf("first number");
    scanf("%d",&x);
    printf("second number");
    scanf("%d",&y);
    printf("third number");
    scanf("%d",&z);

    if(x>y && x>z){
        printf("%d is the largest ODD value mong three digit \n",x);
    }
    if(y>x && y>z){
        printf("%d is the largest ODD value mong three digit \n",y);
    }
    if(z>y && z>x){
        printf("%d is the largest ODD value mong three digit \n",z);
    }
    return 0;
}

```

```
C:\Users\Lenovo\Desktop\1.exe
first number3
second number5
third number7
7 is the largest ODD value mong three digit

-----
Process exited after 7.141 seconds with return value 0
Press any key to continue . . .
```

PROBLEM 06

```
#include<stdio.h>

int main()

{
int x, y, z;
printf("3 sides of triangle\n");
scanf("%d%d%d",&x,&y,&z);
if (x==y&&y==z)
printf("triangle is equilateral");
else if (x==y&&y!=z)
printf("triangle is issocoles");
else if(x==z&&y!=x)
printf("triangle is issocoles");
else if (y==z&&y!=x)
```

```
printf("triangle is issocelles");
```

```
else
```

```
printf("triangle is scalene");}
```

```
#include<stdio.h>
int main()

{
int x, y, z;
printf("3 sides of triangle\n");
scanf("%d%d%d",&x,&y,&z);
if (x==y&&y==z)
printf("triangle is equilateral");
else if (x==y&&y!=z)
printf("triangle is issocelles");
else if (x==z&&y!=x)
printf("triangle is issocelles");
else if (y==z&&y!=x)
printf("triangle is issocelles");
else
printf("triangle is scalene");
}
```

```
C:\Users\Lenovo\Desktop\1.exe
3 sides of triangle
6
6
6
triangle is equilateral
-----
Process exited after 9.494 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Lenovo\Desktop\1.exe
3 sides of triangle
6
6
2
triangle is issocelles
-----
Process exited after 7.358 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Lenovo\Desktop\1.exe
3 sides of triangle
9
3
4
triangle is scalene
-----
Process exited after 10.77 seconds with return value 0
Press any key to continue . . .
```

PROBLEM 07

```
#include<stdio.h>

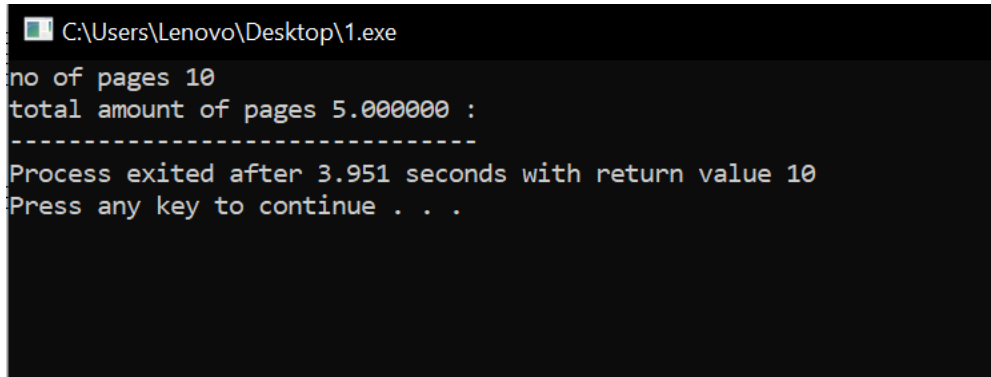
float amountofpages(float x, float result, float finalresult){
    if (x<=10){
        result = (x*0.20)+3.00;
        printf("total amount of pages %f : ",result);
    }
    if (x>10){
        finalresult=((10.0*0.20)+3.00)+((x-10.0)*0.1+3.0);
        printf("total amount of pages %f : ",finalresult);
    }
}

int main()
```

```

{
    float x,result,finalresult;
    printf("no of pages");
    scanf("%f",&x);
    result=amountofpages(x,result,finalresult);
    return result; }

```



```

C:\Users\Lenovo\Desktop\1.exe
no of pages 10
total amount of pages 5.000000 :
-----
Process exited after 3.951 seconds with return value 10
Press any key to continue . . .

```

```

#include<stdio.h>
float amountofpages(float x, float result, float finalresult){
    if (x<=10){
        result = (x*0.20)+3.00;
        printf("total amount of pages %f : ",result);
    }
    if (x>10){
        finalresult=((10.0*0.20)+3.00)+((x-10.0)*0.1+3.0);
        printf("total amount of pages %f : ",finalresult);
    }
}

int main()
{
    float x,result,finalresult;
    printf("no of pages");
    scanf("%f",&x);
    result=amountofpages(x,result,finalresult);
    return result;
}

```